

E-Commerce Sales Data Analytics Report

1. Problem Statement

The objective of this project is to analyze the sales data of an e-commerce business to understand overall business performance. The analysis focuses on identifying sales trends, customer purchasing behavior, product performance, and regional demand. The goal of this analysis is to provide data-driven insights that can help the business improve sales strategies, optimize product offerings, and enhance customer satisfaction.

2. Dataset Overview

The dataset used for this project contains sales transaction data of an e-commerce store.

Key fields in the dataset include:

Order Information

- Order ID
- Order Status

Product Information

- Product
- Category
- Price

Customer Information

- Customer Name
- City

Sales Metrics

- Quantity
- Sales

Transaction Information

- Payment Method

The dataset contains multiple sales records that represent customer purchases across different cities, products, and payment methods.

3. Data Preprocessing Steps

Before creating the dashboards, the dataset was cleaned and prepared using the following preprocessing steps:

- Checked the dataset for duplicate records.
- Verified that all columns had the correct data types.
- Ensured there were no missing or null values in important fields.
- Standardized column names for better readability.
- Verified numerical columns such as Sales, Quantity, and Price.
- Loaded the cleaned dataset into Power BI for visualization.

These preprocessing steps ensured the data was accurate and ready for analysis.

4. Dashboard Pages Created

Three dashboards were created in Power BI to analyze the dataset from different perspectives.

4.1 Dashboard 1 – Sales Overview

Purpose:

To provide a high-level overview of overall sales performance.

Visualizations used:

- KPI Cards (Total Sales, Total Orders, Total Quantity)
- Sales by Category
- Sales by Product
- Quantity by Product
- Category Sales Contribution (Donut Chart)
- Product Sales Distribution (Treemap)

This dashboard helps understand which categories and products contribute the most to overall sales.

4.2 Dashboard 2 – City and Payment Analysis

Purpose:

To analyze regional demand and customer payment preferences.

Visualizations used:

- Orders by City
- Sales by City
- Orders by Payment Method
- Order Status Distribution
- Sales by Payment Method
- Orders by Category

This dashboard highlights how sales vary across cities and how customers prefer to pay for their purchases.

4.3 Dashboard 3 – Customer Insights

Purpose:

To understand customer behavior and identify high-value customers.

Visualizations used:

- Top Customers by Sales
- Orders by Customer
- Customer Sales Contribution (Treemap)
- Sales by Order Status
- Quantity Sold by City
- Sales by Category and Payment Method

This dashboard helps identify loyal customers and analyze purchasing patterns.

5. DAX Measures Used

The following measures were created in Power BI using DAX.

Total Sales

Total revenue generated from all transactions.

Total Sales = SUM(Sales)

Total Orders

Total number of orders placed.

Total Orders = COUNT(Order_ID)

Total Quantity

Total number of items sold.

Total Quantity = SUM(Quantity)

These measures were used in multiple charts and KPI cards across the dashboards.

6. Key Insights and Findings

Based on the analysis, several insights were identified:

- Certain product categories generate higher sales than others.
- Some cities contribute significantly more orders and revenue.
- A small group of customers contributes a large portion of total sales.
- Payment methods such as UPI and card payments are commonly used by customers.
- Completed orders account for the majority of transactions.

These insights help understand business performance and customer behavior.

7. Business Recommendations

Based on the findings from the dashboards, the following recommendations can help improve business performance:

- Focus marketing campaigns on high-performing cities.
- Promote high-selling products to increase revenue.
- Offer loyalty programs for top customers.
- Encourage digital payment methods for faster transactions.
- Analyze low-performing products and adjust pricing or promotions.

Implementing these strategies can help the business improve sales and customer engagement.

8. Conclusion

The E-Commerce Sales Data Analytics project provides valuable insights into sales performance, product demand, and customer behavior. By using Power BI dashboards and data analysis techniques, the business can identify growth opportunities and make better strategic decisions. The interactive dashboards allow stakeholders to explore the data easily and gain meaningful insights to support business growth.